



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VINAYAGA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

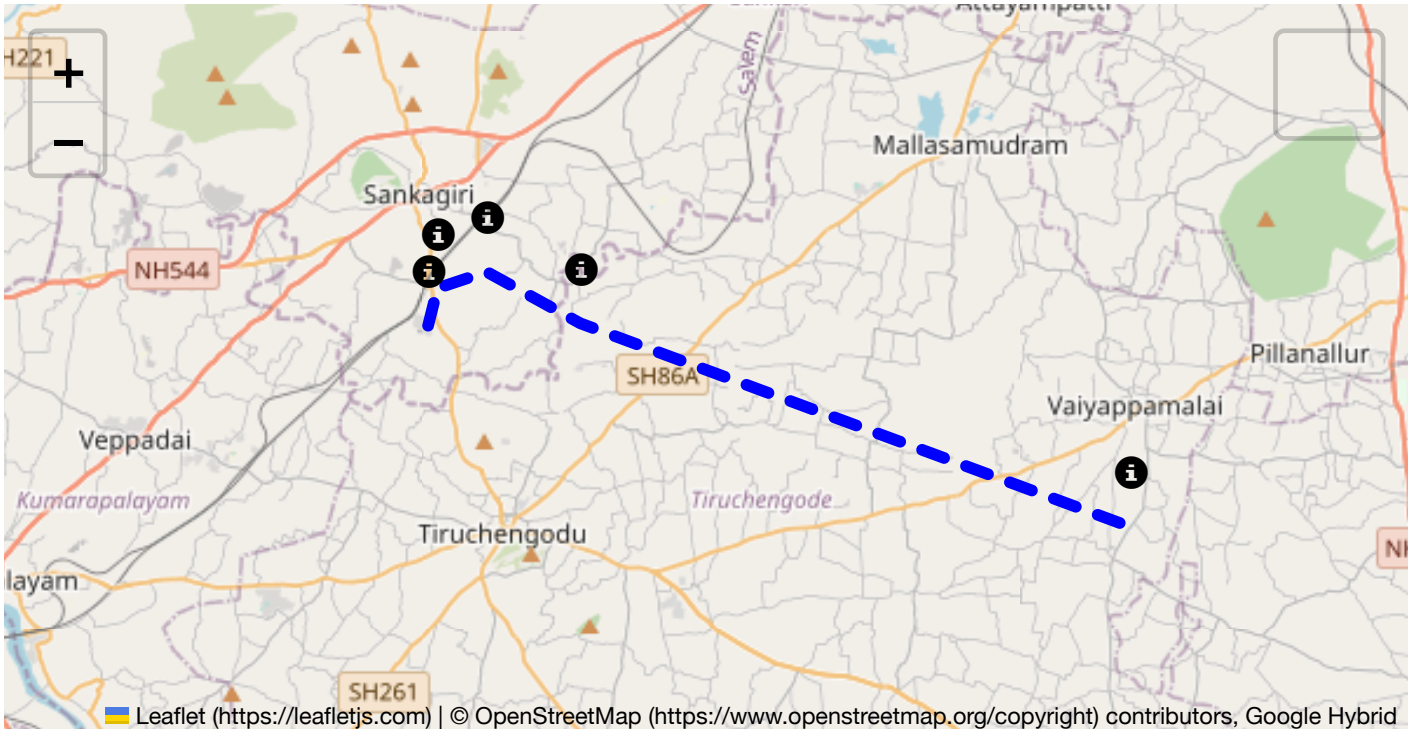
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

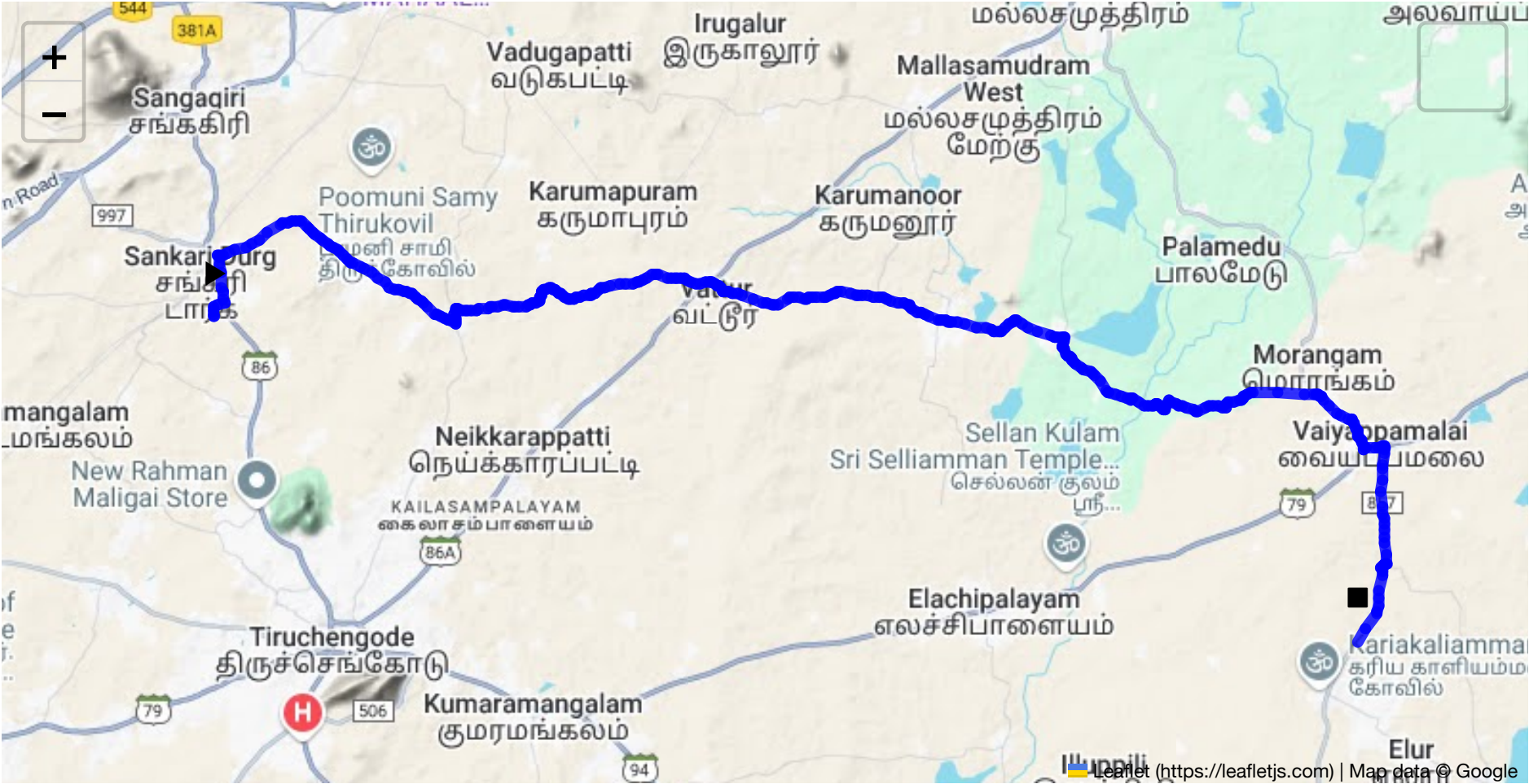
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 32.15 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours

Start: (11.4381, 77.8734)
End: (11.380649, 78.079178)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 93JH+7MC, Kokkalai passes through several smaller towns and rural areas, including CVXG+8MC, FV3R+C64, and Kozhikkalnatham. It spans approximately 32.15 kilometers and is a mix of town roads and state highways, typically taking about 52 minutes for heavy vehicles. Given the rural nature of parts of the route, expect a combination of two-lane roads with variable quality.

Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with heavy rainfall during the monsoon season (June to September). During this time, roads can become slippery, and visibility may be reduced. Additionally, summer months (March to May) can be extremely hot, which may affect vehicle performance.

Traffic Patterns

Traffic congestion is typically moderate, increasing around school zones and marketplaces, especially in Sangagiri. Peak hours usually align with school and office timings, around 8-10 AM and 5-7 PM. Farmers moving goods to market can also create temporary congestion in rural areas.

Road Quality and Infrastructure

The roads are generally well-maintained but can have uneven patches, especially in less urban areas. Some portions may lack proper signage and lighting, especially important for driving at night. Watch for unexpected potholes and narrow stretches in village areas.

Alternative Routes for Emergencies

In case of road blockages or emergencies, a potential diversion could involve taking the main highway back towards Erode before rerouting south towards Kokkalai. This adds distance but utilizes better-quality highways.

Local Regulations on Hazardous Material Transport

Vehicles carrying hazardous materials must adhere to specific timings, usually restricted to avoid peak city traffic. It is essential for drivers to have complete documentation and comply with hazardous substance markings as per local regulations.

Historical Incidents

There have been few reported incidents involving heavy vehicles on this particular route. However, accidents involving smaller transport vehicles and motorcycles are more common, primarily near intersections and market areas.

Environmental Considerations and Sensitive Areas

The route passes through several agricultural and residential areas, necessitating caution to avoid contamination or accidents that could affect local communities. Noise and pollution concerns should also be minimized.

Communication Coverage and Dead Zones

Mobile network coverage is generally good, but there may be occasional dead zones, particularly in more remote areas between towns. It's advisable to have offline navigation tools or physical maps as a backup.

Emergency Response Times

Response times can vary significantly. In urban areas like Sangagiri, expect emergency services to arrive within 15-30 minutes. In rural areas, response times may exceed 30-45 minutes due to road conditions and distance.

Overall Risk Assessment Summary

The primary risks on this route include variable weather conditions, potential road quality issues, and moderate traffic congestion. While the route itself is not typically hazardous, careful driving is necessary, especially considering the transport of hazardous materials. Effective pre-trip planning, adherence to local regulations, and situational awareness can greatly reduce risks. Always prepare for worst-case scenarios with alternative routes and emergency contacts at the ready.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
4	Turn	Medium	11.44194, 77.90923	30 KM/Hr
5	Blind Spot	Blind Spot	11.43673, 77.91703	10 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.43931, 77.92291	30 KM/Hr
8	Turn	Medium	11.43982, 77.93018	30 KM/Hr
9	Turn	Medium	11.44033, 77.93076	30 KM/Hr
10	Turn	Medium	11.44071, 77.93211	30 KM/Hr
11	Turn	Medium	11.44096, 77.93223	30 KM/Hr
12	Turn	Medium	11.44135, 77.93220	30 KM/Hr
13	Turn	Medium	11.44175, 77.93249	30 KM/Hr
14	Turn	High	11.44248, 77.93247	15 KM/Hr
15	Turn	Medium	11.44547, 77.95309	30 KM/Hr
16	Turn	Medium	11.44370, 77.96020	30 KM/Hr
17	Turn	Medium	11.44394, 77.96273	30 KM/Hr
18	Turn	High	11.44387, 77.96283	15 KM/Hr
19	Turn	High	11.44395, 77.96298	15 KM/Hr
20	Turn	Medium	11.44302, 77.96480	30 KM/Hr
21	Turn	High	11.44243, 77.96500	15 KM/Hr
22	Turn	Medium	11.44011, 77.97521	30 KM/Hr
23	Turn	Medium	11.44158, 77.98455	30 KM/Hr
24	Turn	Medium	11.43528, 78.01534	30 KM/Hr
25	Turn	Medium	11.43720, 78.01774	30 KM/Hr
26	Turn	High	11.43441, 78.02649	15 KM/Hr
27	Turn	Medium	11.43427, 78.02658	30 KM/Hr
28	Turn	Medium	11.43260, 78.02641	30 KM/Hr
29	Turn	Medium	11.43135, 78.02685	30 KM/Hr
30	Turn	Medium	11.42462, 78.03450	30 KM/Hr
31	Turn	High	11.42324, 78.03848	15 KM/Hr
32	Turn	Medium	11.42336, 78.03865	30 KM/Hr
33	Turn	Medium	11.42360, 78.03876	30 KM/Hr
34	Turn	High	11.42366, 78.03889	15 KM/Hr
35	Turn	High	11.42216, 78.04283	15 KM/Hr
36	Turn	High	11.42261, 78.04307	15 KM/Hr
37	Turn	Medium	11.42260, 78.04326	30 KM/Hr
38	Turn	High	11.42151, 78.04462	15 KM/Hr
39	Turn	Medium	11.42157, 78.04478	30 KM/Hr
40	Blind Spot	Blind Spot	11.42318, 78.04539	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
41	Turn	Medium	11.42120, 78.05015	30 KM/Hr
42	Turn	High	11.42195, 78.05572	15 KM/Hr
43	Turn	High	11.42259, 78.05580	15 KM/Hr
44	Turn	Medium	11.42326, 78.05838	30 KM/Hr
45	Turn	High	11.42430, 78.07221	15 KM/Hr
46	Blind Spot	Blind Spot	11.41453, 78.08062	10 KM/Hr
47	Turn	Medium	11.41470, 78.08074	30 KM/Hr
48	Blind Spot	Blind Spot	11.41508, 78.08433	10 KM/Hr
49	Turn	High	11.39425, 78.08455	15 KM/Hr
50	Turn	Medium	11.39377, 78.08379	30 KM/Hr
51	Turn	Medium	11.39350, 78.08368	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Sri Vinayaga Matriculation Higher Secondary School	11.4233682, 78.0736517	30 km/h	Medium
1	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
2	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44194, 77.90923



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43931, 77.92291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43982, 77.93018



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44033, 77.93076



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44071, 77.93211



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44096, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44135, 77.93220



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44175, 77.93249



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44248, 77.93247



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44547, 77.95309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44370, 77.96020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44394, 77.96273



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44387, 77.96283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44395, 77.96298



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44302, 77.96480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.96500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44011, 77.97521



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44158, 77.98455



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43528, 78.01534



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43720, 78.01774



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43441, 78.02649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43427, 78.02658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43260, 78.02641



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43135, 78.02685



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42462, 78.03450



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42324, 78.03848



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42336, 78.03865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42360, 78.03876



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42366, 78.03889



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42216, 78.04283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42261, 78.04307



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42260, 78.04326



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42151, 78.04462



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42157, 78.04478



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42318, 78.04539



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42120, 78.05015



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42195, 78.05572



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42259, 78.05580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42326, 78.05838



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42430, 78.07221



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41470, 78.08074



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41508, 78.08433



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39425, 78.08455



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39377, 78.08379



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39350, 78.08368

